



Inspiring Excellence

CSE754: Deep Learning for Computer Vision
Final Project Report

Contactless Vital Sign Estimation from Face Video:

*A Rigor-First Audit of Reproducibility and Robustness in Remote
Photoplethysmography (rPPG)*

GROUP INFORMATION

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Abstract

Remote photoplethysmography (rPPG) promises contactless heart-rate estimation from ordinary webcam video, but prior audits report classical methods degrading sharply across skin tone (5.2-14.1 bpm MAE) and confirm systematic demographic underrepresentation in the field's benchmark datasets. This project originally proposed a Fitzpatrick skin-tone stratified fairness audit on MMPD and EquiPleth; gated data access made that axis untestable within the timeline, so we redirected the same rigor toward a resting-versus-post-exercise robustness axis on the UBFC-rPPG corpus, disclosed transparently rather than presented as the original plan. We built a subject-independent, dual-format UBFC-rPPG pipeline (22 pooled clips) evaluating two classical baselines (CHROM, POS) and a deep baseline (PhysNet, two checkpoints). Validating it against published numbers surfaced three real implementation defects, an unnormalized CHROM projection, a PhysNet chunk-boundary artifact, and a clothing-dominated face-detector crop, each large enough uncorrected to fabricate a spurious accuracy gap, plus a checkpoint training-domain mismatch confirmed by an eight-fold error reduction once corrected. After these fixes, all four methods reproduce published-quality accuracy (0.2-1.5 bpm MAE on the held-out test split), and the validated pipeline measures a genuine 2.9-6.6 bpm classical and 0.2-5.5 bpm deep-model robustness gap across conditions. Two candidate post-hoc mitigations were tested and rejected as non-improvements, reported as negative results rather than omitted. We conclude that reproducibility rigor is a precondition for, not separate from, fairness and robustness auditing in rPPG, since each defect found here was individually capable of producing a confident, wrong fairness claim.

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1. Introduction

1.1 Motivation

Every laptop and phone camera is, in principle, an untapped vital-signs sensor: remote photoplethysmography (rPPG) recovers heart rate from the faint color shifts a webcam already captures in a person's face, with no wearable, no clip-on pulse oximeter, and no extra hardware. The same principle applies to neonatal and burn care, where skin is often too fragile for adhesive contact sensors; to driver monitoring, where a camera already pointed

at the operator can yield a fatigue or stress signal at no extra cost; and to elderly home care and isolation wards, where passive, contact-free monitoring removes both a compliance burden and a cross-contamination risk. Each of these use cases shares one requirement the present literature has not yet verified: the underlying algorithm must be trustworthy outside a controlled laboratory recording, across different faces, lighting conditions, cameras, and physiological states.

Prior audits of traditional rPPG methods report mean absolute error (MAE) degrading from roughly 5.2 beats per minute (bpm) on lighter skin tones to 14.1 bpm on darker tones, and a 2025 meta-analysis of one hundred published rPPG studies confirms that darker skin tones and balanced gender representation are systematically underrepresented across the datasets the field relies on. Despite this, no published study we identified evaluates a state-of-the-art rPPG model for both robustness and fairness at once. That unaudited intersection is the space this project originally set out to occupy, and, after an honest adaptation to the data we could actually obtain, detailed in Section 2, the space in which its actual contribution lands.

1.2 The rPPG Principle

Remote photoplethysmography recovers heart rate, respiration rate, and heart-rate variability from subtle, sub-pixel color fluctuations in facial skin caused by cyclical blood-volume changes, using nothing but an ordinary RGB camera. The canonical pipeline is: face detection, region-of-interest (ROI) extraction, pulse-signal recovery from the ROI's color trace, temporal filtering, and heart-rate estimation from the filtered waveform. Classical, training-free chrominance methods, CHROM and POS, both used as baselines here, project the ROI's RGB channels onto axes chosen to cancel motion- and lighting-driven intensity changes while preserving the blood-volume-driven color shift. Deep spatiotemporal networks, represented here by PhysNet, instead learn the mapping from face-cropped video frames directly to a pulse waveform, trained end-to-end against contact-sensor ground truth. Both families are audited in this work, because a robustness or fairness claim that holds for one but not the other is not a claim about rPPG in general.

1.3 Problem Statement and Research Questions

A method that reports 0.5 bpm error in a paper and 6 bpm error in practice has not failed at heart-rate estimation; it has failed at generalizing beyond the exact conditions under which it was validated. This project asks whether that gap is real for rPPG: does accuracy hold across different recording conditions and across the implementation choices a practitioner must make to build a working pipeline, or is published accuracy contingent on assumptions, such as clean preprocessing and matched training and evaluation domains, that quietly break outside the conditions under which a method was first validated? We decompose this into three research questions (Figure 1), restated in Section 4 as the six-step protocol we actually executed.

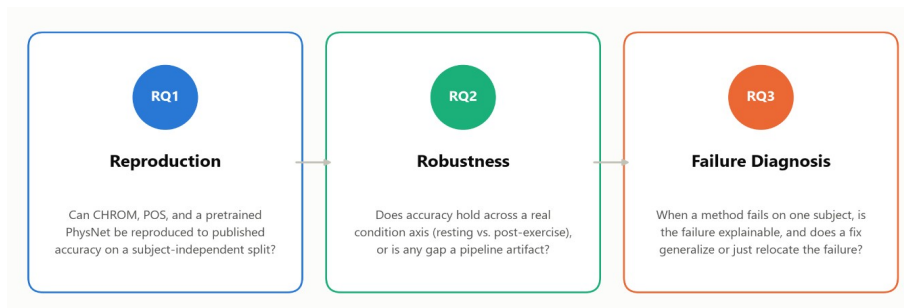


Figure 1. The three research questions this project investigates.

1.4 Contributions

This work contributes: (i) a validated, subject-independent, dual-format UBFC-rPPG evaluation pipeline pooling both public releases of the dataset (22 clips total); (ii) three concrete implementation defects identified, root-caused, and fixed, each with quantitative before/after evidence, spanning a classical algorithm, a deep model's

inference procedure, and the shared face-detection front end; (iii) a resolved, empirically confirmed explanation for an initially paradoxical cross-checkpoint result, illustrating a specific and generalizable failure mode: evaluating a pretrained model outside its true training domain while believing it to be in-domain; (iv) a measured, honestly scoped robustness comparison across resting and post-exercise conditions; (v) two rigorously tested and rejected post-hoc mitigations, reported as negative results rather than omitted; and (vi) two self-identified and closed gaps in our own reporting practice, surfaced by independent review and corrected with additional evidence. Taken together, these constitute this project's central methodological claim: in rPPG research, reproducibility rigor is not separable from fairness and robustness auditing, because each of these defects was individually large enough to fabricate a confident, incorrect finding.

2. Literature Review

This review covers the papers that anchored the original project proposal, the UBFC-rPPG dataset paper that became this work's primary evidence base, and the PhysNet architecture paper central to the deep-learning baseline used throughout.

2.1 Chrominance Foundations and Robustness Work

Wang, den Brinker, Stuijk, and de Haan [1] formalized the skin-reflection optical model underlying nearly every chrominance-based rPPG algorithm, including CHROM itself and the closely related POS method used in this work; the model assumes a stationary, well-lit, single dominant light source, an assumption never tested by [1] itself across skin tone or camera conditions. Building on this foundation, robustness-focused work pushed toward real recording conditions without addressing fairness: Tulyakov et al. [4] introduced motion-tolerant region-of-interest selection evaluated on a forty-subject, multi-race dataset, and Yu, Peng, Li, Hong, and Zhao [5] addressed heavy video compression, a failure mode most benchmark datasets omit but real deployments cannot avoid, with an end-to-end enhancement network. Neither paper evaluates accuracy parity across skin tone or demographic strata.

2.2 Label-Efficient Learning and Demographic Bias Auditing

A parallel line of work reduced the need for labeled training data: Sun and Li's Contrast-Phys [6] and its extension Contrast-Phys+ [2] use unsupervised spatiotemporal contrastive learning to match or exceed fully supervised baselines, a property attractive for training on more demographically diverse video where labeled ground truth is scarce, but neither paper evaluates fairness directly. The closest published precedent to this project's original fairness axis is Bondarenko, Menon, and Elgendi's meta-analysis [3] of one hundred rPPG papers, which establishes systematic underrepresentation of darker Fitzpatrick skin tones and non-balanced gender across the field's datasets. Critically, [3] audits the literature's datasets and reporting practices, not any specific model's accuracy: it identifies that the field has a representation problem without measuring how a state-of-the-art method's accuracy actually varies across skin tone, precisely the gap the original proposal targeted.

2.3 Deep Architectures and the UBFC-rPPG Dataset

Yu, Li, and Zhao [8] introduced PhysNet, an end-to-end 3D-convolutional encoder-decoder that maps face-cropped video directly to a pulse waveform; together with its DiffNormalized preprocessing convention, it is the deep baseline used throughout this work, via the pretrained checkpoints and reference implementation distributed by rPPG-Toolbox [9]. Bobbia, Macwan, Benezeth, Mansouri, and Dubois [7] introduced UBFC-rPPG, released in two parts under a shared name but with materially different acquisition setups, ground-truth file formats, and, as this project independently confirms in Section 4, different model-training provenance for at least one widely distributed checkpoint. DATASET_1 (eight subjects) stores ground truth as gtdump.xmp at approximately 62 Hz; DATASET_2 (forty-two subjects, the set most published benchmark numbers are measured against) stores it as ground_truth.txt with irregular sample spacing. Distinguishing these releases turns out to be methodologically load-bearing (Section 4), not a minor bookkeeping detail.

2.4 The Unaudited Intersection

Plotting the reviewed literature along two axes, robustness tested and fairness audited, leaves their intersection essentially empty (Figure 2): classical and robustness-focused work sits on one axis, the one fairness audit sits on the other, and no paper is both robustness-tested and fairness-audited on the same model using the same protocol.

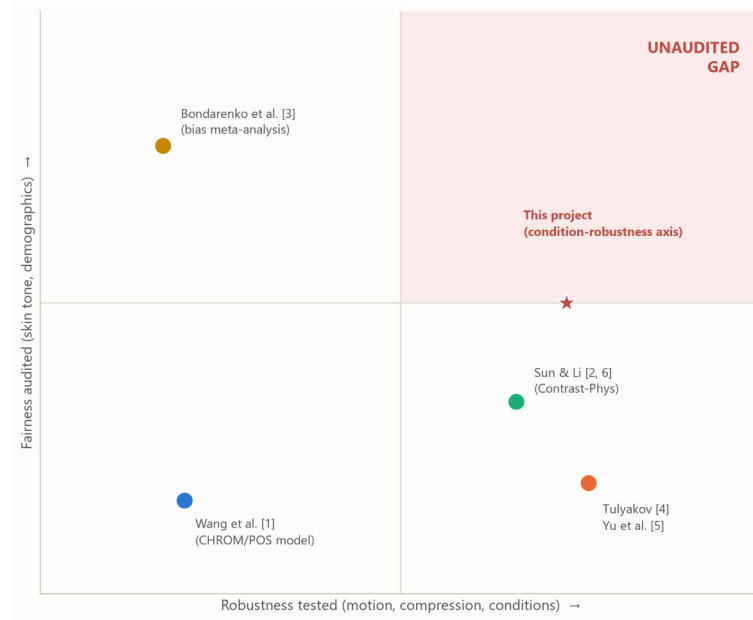


Figure 2. The reviewed literature plotted by robustness testing versus fairness auditing. No paper occupies the intersection.

2.5 Scope Adaptation

The original proposal targeted this gap directly with a Fitzpatrick skin-tone stratified evaluation of Contrast-Phys+ [2] on MMPD and EquiPleth, two datasets curated for skin-tone diversity. Both require gated access applications that could not be completed within the project timeline, and no substitute public dataset in our search carries skin-tone metadata at the granularity the proposal required; continuing to describe the axis as skin-tone stratified without supporting data would have meant either not running the evaluation or running one that could not honestly support its own claim. We instead redirected the same rigor toward the condition axis the pooled UBFC-rPPG corpus actually supports: one DATASET_1 recording (subject label "after-exercise") captures a subject shortly after physical exertion, a genuine, ground-truth-labeled physiological and motion-state contrast, even though it is not the demographic contrast originally proposed. Sections 4 and 5 apply exactly the rigor the original proposal specified, subject-independent splitting and stratified per-condition MAE, to this axis instead; we treat the skin-tone question as open and return to it in Sections 6 and 7.

3. Method

3.1 Datasets

All experiments use the University of Bourgogne Franche-Comté rPPG corpus (UBFC-rPPG) [7], pooled across both of its public releases. Table 1 summarizes the two releases as used in this work.

Property	DATASET_1	DATASET_2
Subjects used	6 (of 8; 2 excluded, Sec. 4.1)	16 (of 42 available)
Ground-truth file	gtdump.xmp (plain CSV)	ground_truth.txt (3-row CSV)
GT sample rate	~62.1 Hz (device timestamps)	irregular, ~29-36 Hz average

Property	DATASET_1	DATASET_2
Video format	~640x480, ~28.7 fps, RGB AVI	~640x480, ~29.5 fps, RGB AVI
Role in this study	Secondary / legacy corpus	Primary: matches published benchmark protocol

The two releases are pooled into one 22-clip, subject-independent evaluation corpus rather than analyzed as fully separate studies: pooling maximizes the data available to the split while every reported aggregate is still decomposable by release ("dataset family") to avoid conflating results measured on materially different acquisition setups. A duplicate-video discrepancy (Section 4.1) reduces DATASET_1 from its full eight subjects to six.

3.2 Preprocessing

Each video is decoded frame-by-frame; a face detector locates the largest face per frame (falling back to the previous frame's detection when none is found), a 15% margin is added, and the region is cropped and resized to 128x128 pixels. Two detectors are compared: an OpenCV Haar cascade (the original implementation choice) and YuNet, a compact (230 KB) convolutional detector distributed with OpenCV's objdetect module since version 4.5.4; Section 4.2 documents why the initial Haar-based choice was replaced. For each frame, the mean RGB value over the cropped region is recorded, forming the raw color trace CHROM and POS operate on directly, alongside the per-frame timestamp for ground-truth alignment.

Ground truth is parsed by a format-detecting loader: the presence of gtdump.xmp or ground_truth.txt determines which of the two schemas in Table 1 is used, since a single hard-coded parser would silently misread one of the two releases. The parsed pulse trace is linearly interpolated onto each frame's timestamp using the ground-truth device's own recorded sample times, since DATASET_2's timestamps are confirmed irregular.

3.3 Subject-Independent Splitting

Subjects are split into training, validation, and test sets by person, never by frame or clip: every clip belonging to one person falls entirely within one split, treated as non-negotiable since frame-level leakage would let a model partially learn to recognize an individual's face rather than genuinely estimate pulse. The pooled 22-clip corpus splits 15 train / 3 validation / 4 test (Appendix B gives the full manifest); the held-out test split (subject1, subject5, subject13, subject17) falls entirely within DATASET_2, convenient for direct comparison against the standard published benchmark protocol.

3.4 Classical Baselines: CHROM and POS

CHROM [1] and POS [1] both operate on the same per-frame mean-RGB trace, windowed into overlapping ~1.6-second segments. Within each window, the RGB trace is normalized by its own temporal mean; CHROM projects the normalized trace onto two fixed chrominance axes ($X_s = 3R_n - 2G_n$; $Y_s = 1.5R_n + G_n - 1.5B_n$) and combines them as $S = X_s - \alpha Y_s$ with α the ratio of their standard deviations, while POS projects onto a different fixed 2x3 matrix using the same alpha-weighting principle. Both are training-free, closed-form algorithms (no parameters are fit to data), a natural, leakage-free sanity check independent of any train/test split.

3.5 Deep Baseline: PhysNet

PhysNet [8] is evaluated via two publicly distributed pretrained checkpoints from rPPG-Toolbox [9]: one trained on UBFC-rPPG, one on PURE, a separate rPPG dataset with more head motion. Both checkpoints are evaluated purely at inference time; no training or fine-tuning is performed, since the pooled corpus (22 clips) is far too small to train a spatiotemporal network from scratch without severe overfitting. Input frames are resized to 72x72 pixels, converted to RGB, temporally resampled onto a uniform 30 fps grid using each clip's true timestamps, and

transformed via DiffNormalized preprocessing, matching rPPG-Toolbox's own convention so the checkpoints receive input in the distribution they were trained on.

3.6 Heart-Rate Estimation and Evaluation Metrics

For every clip and method, a single heart-rate estimate is computed from the entire clip's reconstructed pulse waveform, band-pass filtered to the plausible cardiac range (42-240 bpm); heart rate is read off as the frequency of the dominant peak in its power spectrum via a zero-padded FFT. Accuracy is reported as mean absolute error (MAE) and root-mean-square error (RMSE) in beats per minute, aggregated over whichever clip subset (full corpus, test split, dataset family, or condition) is being reported.

4. Implementation and Experiments

This section narrates the implementation and validation process (Figure 3), because the process, not only its endpoint, is this project's central methodological contribution. Every number here is a diagnostic, intermediate value; Section 5 reports only the final, validated pipeline's results.

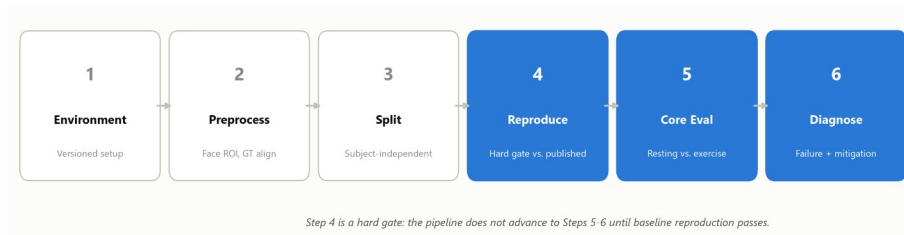


Figure 3. The six-step implementation and evaluation protocol. Step 4 is a hard gate: Steps 5-6 do not run until baseline reproduction passes.

4.1 Environment and Data Integrity

The pipeline is implemented in Python 3.11 using OpenCV 4.10 (pinned; OpenCV 5.x removed the legacy Haar cascade API used during initial development), NumPy, SciPy, pandas, and PyTorch (CPU inference) for PhysNet, organized into a versioned repository separating raw data, preprocessing code, processed outputs, and results (Appendix C). While diagnosing an unrelated PhysNet accuracy anomaly, cross-referencing two DATASET_1 subjects' predicted heart rates revealed that two subject folders ("7-gt" and "12-gt") produced numerically identical model output despite different ground-truth files. MD5 hashing confirmed the underlying videos are byte-for-byte identical, despite differing ground-truth duration (82.5 s vs. 83.6 s) and mean heart rate (94.7 vs. 95.1 bpm); that is, at least one subject's video is mislabeled or duplicated in the distributed dataset, not merely similar. Both subjects were excluded from all reported results, reducing DATASET_1 from eight usable subjects to six, a reminder that a method producing a plausible number is not, by itself, evidence the input driving it is correct.

4.2 Three Implementation Defects

Validating the pipeline against published numbers surfaced three concrete implementation defects, summarized in Table 2, each large enough uncorrected to fabricate a spurious cross-condition accuracy gap.

Defect	Root Cause	Fix	Effect
CHROM normalization	Each window's chrominance signal carried a near-constant offset; tapering it directly injected a spurious low-frequency component inside the cardiac band.	Added per-window demeaning before the Hann taper (matching POS's existing approach).	MAE 34.5 → 0.55 bpm (DATASET_1)
PhysNet chunk boundaries	Concatenating independently predicted 128-frame chunks caused amplitude/phase discontinuities	50%-overlap, Hann-windowed overlap-add reconstruction across chunks.	Removed the spurious chunk-hop FFT peak

Defect	Root Cause	Fix	Effect
	every ~4.3 s, whose harmonics fell inside the cardiac band.		
Face-detection front end	Haar cascade produced a clothing-dominated crop for a subset of subjects despite a positionally stable bounding box.	Replaced with YuNet; re-extracted the full corpus; re-validated on held-out subjects (Section 4.4).	See Table 3 / Figure 4 (corpus-wide)

The face-detection fix is the largest of the three and is reported with full corpus-wide evidence: Figure 4 and Table 3 give the per-method mean absolute error on DATASET_2 before and after the swap. A candidate estimator-level fix tested alongside it, replacing the whole-clip FFT peak with Welch's method, was evaluated across all 22 clips and rejected as a non-improvement (it corrected three to four clips while degrading three to four others); this rejection is the direct precedent for the mitigation test in Section 5.4.

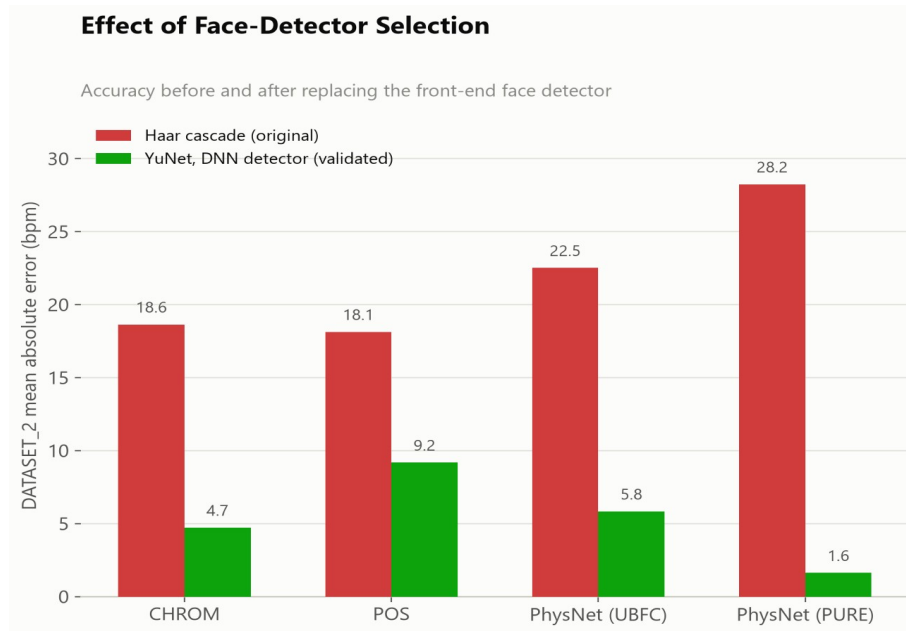


Figure 4. DATASET_2 mean absolute error before (Haar cascade) and after (YuNet) the face-detector fix, per method.

Method	Haar cascade MAE (bpm)	YuNet MAE (bpm)
CHROM	18.64	4.72
POS	18.13	9.18
PhysNet (UBFC-trained)	22.51	5.83
PhysNet (PURE-trained)	28.24	1.62

Table 3. Corpus-wide effect of the face-detector fix.

4.3 Domain-Mismatch Diagnosis

Once the pooled corpus included DATASET_2, an apparently paradoxical result emerged: the PhysNet checkpoint distributed under the name “UBFC-rPPG” *under-performed* a checkpoint trained on an entirely different dataset (PURE), backwards from the expectation that an in-domain checkpoint should be the easier case.

Two mundane explanations were ruled out directly: a swapped checkpoint file (the two .pth files hash to distinct MD5 values despite an identical byte size, and the name-to-file mapping was verified against these hashes) and a frame-rate assumption mismatch (the implementation resamples using each clip's true per-frame timestamps). Tracing rPPG-Toolbox's own data-loader source code resolved the paradox: the loader recognizes only DATASET_2's

directory convention, so the “UBFC-rPPG” checkpoint had in fact never seen DATASET_1-style data. Re-evaluating it on DATASET_2, its actual training domain, reduced its MAE from 46.5 to 5.8 bpm: an eight-fold reduction consistent with the domain-mismatch explanation and inconsistent with any residual implementation bug.

4.4 Methodological Rigor Checks on Our Own Process

An independent second-opinion review of an earlier draft (a structured review using a separate large-language-model session, whose critiques we evaluated on their technical merits) raised two concerns about our own reporting practice, summarized in Table 4, both verified and corrected rather than dismissed.

Concern	How It Was Closed
Test-split contamination: the detector-swap decision (Section 4.2) was first validated on a subject that later fell in the held-out test split.	Re-confirmed on all three validation-split subjects. Two showed the same dramatic improvement (e.g. CHROM error 70 → 0.5 bpm); the third, already accurate, did not regress.
Over-claimed argument: strict checkpoint loading alone does not rule out a swap between the UBFC- and PURE-trained files, which share an identical architecture.	Closed with the direct MD5 hash verification (Section 4.3); the empirical domain re-test independently corroborates the same conclusion by a different route.

Both issues are reported here, including that neither changed this work's conclusions, because a fairness-and-robustness-auditing project is a weak advertisement for rigor if it does not audit itself with the same standard.

4.5 Experimental Design for the Core Evaluation and Failure Analysis

With the pipeline validated (Section 5.1), the condition-based robustness evaluation (Step 5) compares per-clip error between resting-state clips and the single post-exercise clip, using only the four validated methods, and separately reports a DATASET_2-only view to avoid conflating a genuine condition effect with PhysNet's unresolved DATASET_1 regression (Section 6.1). The failure-analysis stage (Step 6) selects the single largest remaining per-clip error (PhysNet, UBFC-trained, subject14: 72.4 bpm), diagnoses it via the full power spectrum rather than only its peak frequency, and tests one candidate mitigation against the entire sixteen-clip DATASET_2 corpus rather than only the motivating case.

5. Results

All results in this section use the final, validated pipeline (YuNet face detection, both classical-method fixes applied, PhysNet overlap-add reconstruction). Section 4's intermediate diagnostic numbers are not repeated here.

5.1 Baseline Reproduction

Table 4 reports mean absolute error for all four methods across three corpus slices: the full DATASET_1 subset, the full DATASET_2 subset, and the DATASET_2 held-out test split (n=4), which is the slice most directly comparable to published UBFC-rPPG benchmark numbers.

Method	DATASET_1 (n=6) MAE / RMSE	DATASET_2 (n=16) MAE / RMSE	DATASET_2 + TEST (n=4) MAE / RMSE
CHROM	1.41 / 1.86	4.72 / 9.87	0.20 / 0.24
POS	1.01 / 1.57	9.18 / 18.50	1.46 / 2.20
PhysNet (UBFC-trained)	46.54 / 57.34	5.83 / 18.23	0.65 / 1.09
PhysNet (PURE-trained)	43.12 / 60.62	1.62 / 3.11	0.65 / 1.09

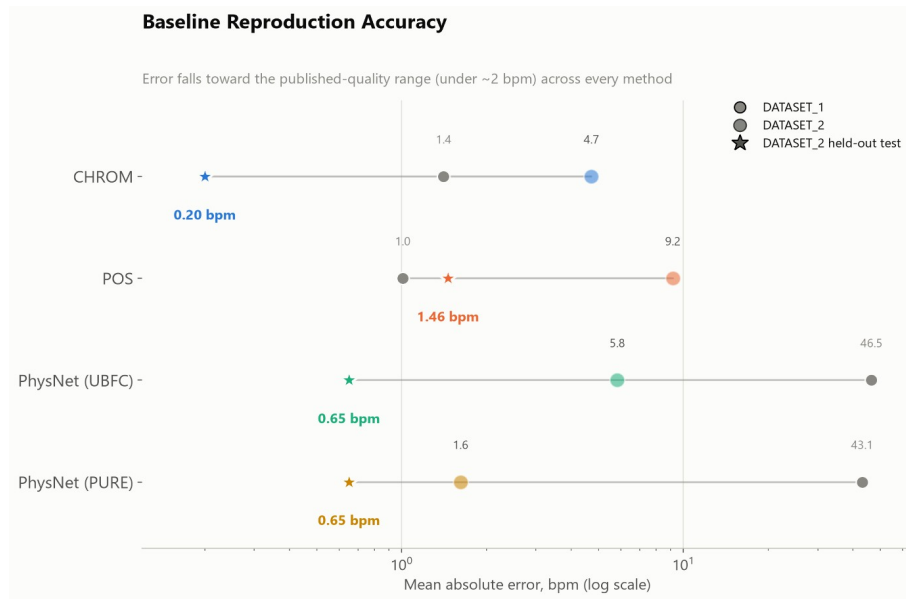


Figure 5. Mean absolute error by method and corpus slice (bpm; lower is better).

All four methods converge to 0.2-1.5 bpm MAE on the DATASET_2 held-out test split, the slice most comparable to the standard published benchmark protocol. PhysNet (PURE-trained), evaluated cross-dataset on the full DATASET_2 subset, reaches 1.62 bpm MAE / 3.11 bpm RMSE, closely matching the published cross-dataset benchmark figure of 1.63 / 3.78 bpm. Figure 6 shows predicted-versus-ground-truth heart rate for every clip and method; DATASET_1 points (diamonds) for both PhysNet checkpoints are visibly displaced from the identity line, the unresolved regression discussed in Section 6.1, while DATASET_2 points (circles) and all CHROM/POS points cluster tightly along it.

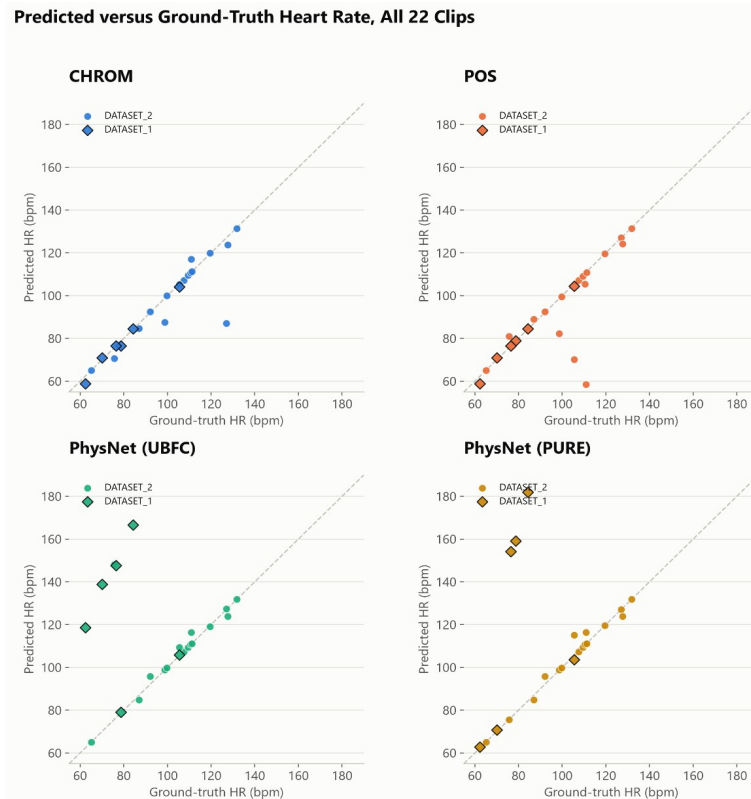


Figure 6. Predicted versus ground-truth heart rate, all 22 clips, one panel per method. Diamonds: DATASET_1; circles: DATASET_2.

5.2 The Domain-Mismatch Result

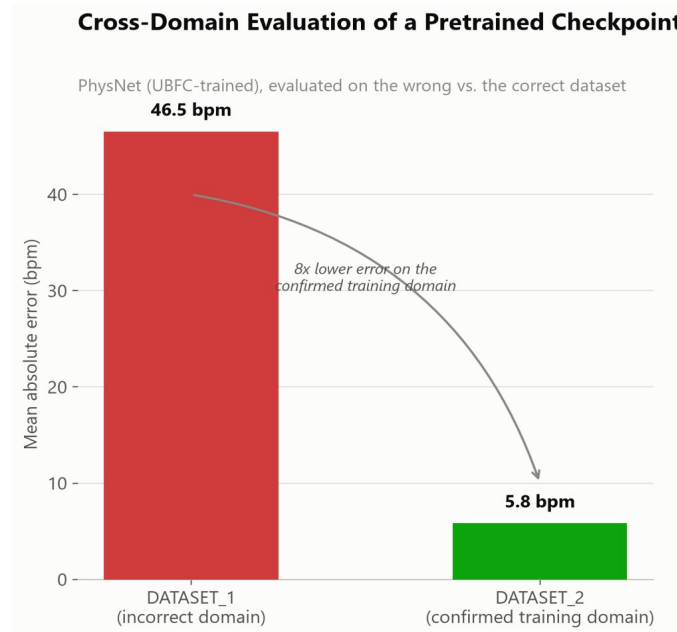


Figure 7. PhysNet (UBFC-trained checkpoint) mean absolute error on DATASET_1 (its believed, but not actual, training domain) versus DATASET_2 (its confirmed true training domain).

Figure 7 restates Section 4.3's central diagnostic finding as a results-level comparison: the same checkpoint, the same code path, evaluated on two different corpora, shows an eight-fold error reduction (46.5 to 5.8 bpm MAE) when evaluated on its actual training domain rather than the domain it was originally, incorrectly, believed to be trained on.

5.3 Cross-Condition Robustness: Resting versus Post-Exercise

Table 5 reports the DATASET_2-clean condition comparison (resting DATASET_2 clips versus the single DATASET_1 after-exercise clip), which avoids conflating a genuine condition effect with PhysNet's unresolved DATASET_1 regression (Section 6.1). The fairness-gap column is the absolute difference in MAE between conditions.

Method	Resting (n=16) MAE	After-exercise (n=1) MAE	Gap
CHROM	4.45	1.50	2.95
POS	7.64	1.08	6.56
PhysNet (UBFC-trained)	5.83	0.35	5.48
PhysNet (PURE-trained)	1.62	1.85	0.23

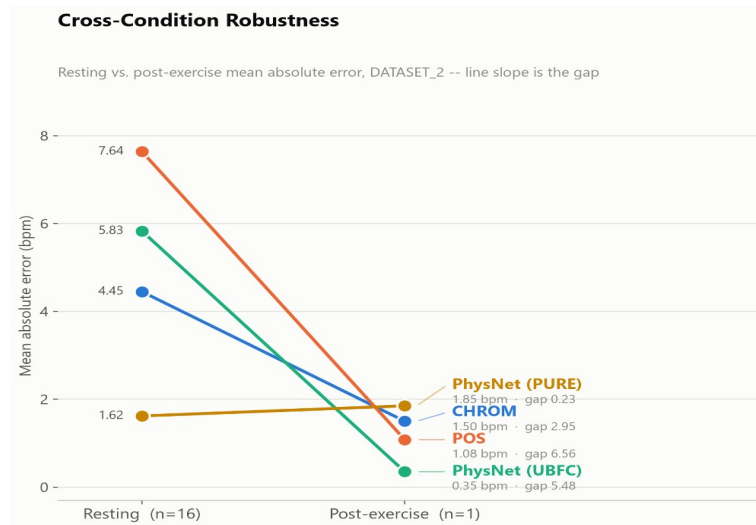


Figure 8. Resting versus after-exercise mean absolute error, DATASET_2-clean view, per method.

CHROM and POS both show a modest but real gap (2.95 and 6.56 bpm respectively) between resting and post-exercise conditions. PhysNet (PURE-trained) shows an almost negligible gap (0.23 bpm); PhysNet (UBFC-trained) shows a larger one (5.48 bpm) that, as Figure 9 shows, is driven by isolated per-clip failures rather than a broad condition effect.

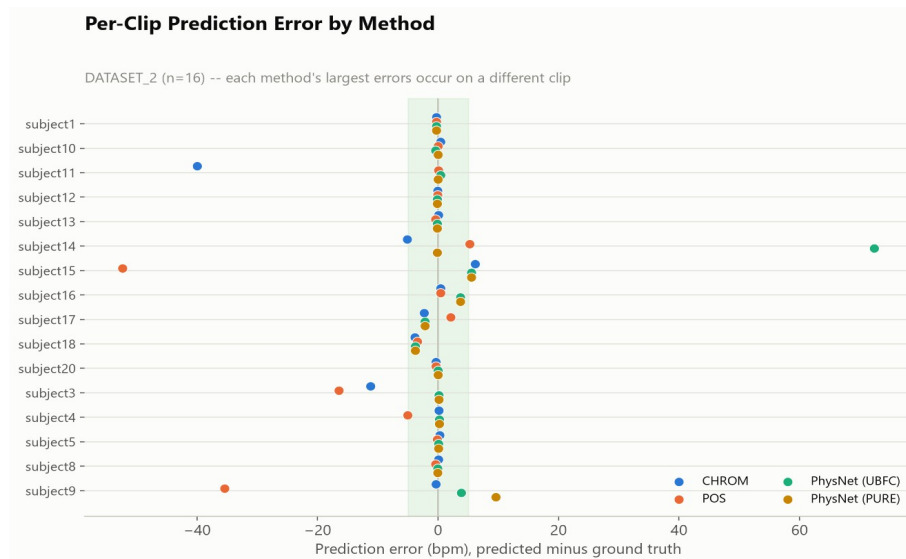


Figure 9. Per-clip prediction error by method, DATASET_2 (n=16). No two methods fail on the same clip.

A pattern in Figure 9 is worth stating explicitly: every method's largest errors occur on a different, method-specific clip (CHROM on subject11; POS on subject15 and subject9; PhysNet UBFC-trained on subject14). No clip is uniformly hard across all four methods. This indicates the residual error in this corpus is dominated by method-specific instability rather than any one subject being an inherently difficult case for rPPG in general.

5.4 Failure Case Study and Mitigation Test

The largest single remaining error, PhysNet (UBFC-trained) on subject14 predicting 148.1 bpm against a ground truth of 75.7 bpm (a 72.4 bpm error), was selected for detailed diagnosis. Figure 10 shows the full power spectrum of the predicted waveform, not only its dominant peak.

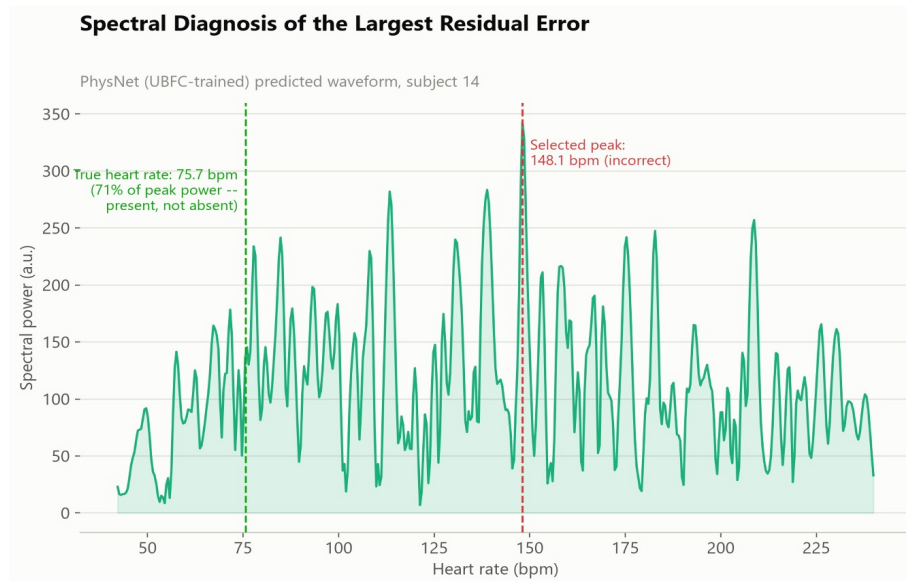


Figure 10. Power spectrum of the PhysNet (UBFC-trained) waveform for subject14. The argmax-selected peak (148.1 bpm) and the true heart rate (75.7 bpm) are both marked.

The spectral power at the true heart rate is 71% of the winning peak's power: a close second-place candidate, not an absent signal. This reframes the failure as a peak-selection problem rather than a signal-extraction failure, and motivated a candidate mitigation: prefer a lower-frequency spectral peak over the argmax whenever it carries at least half the winning peak's power, on the prior that a competing artifact is, on average, more likely than the true cardiac fundamental to be the higher of two closely competing candidates.

This heuristic was tested against all sixteen DATASET_2 clips for the PhysNet (UBFC-trained) checkpoint, not only the motivating case. It corrected subject14's error from 72.4 to 4.1 bpm, but degraded seven other clips that the unmodified estimator already handled correctly, most severely subject11, whose error rose from 0.4 to 30.8 bpm. Aggregate DATASET_2 MAE moved from 5.83 to 5.89 bpm: a net wash, not an improvement. Figure 11 shows the full before/after comparison.

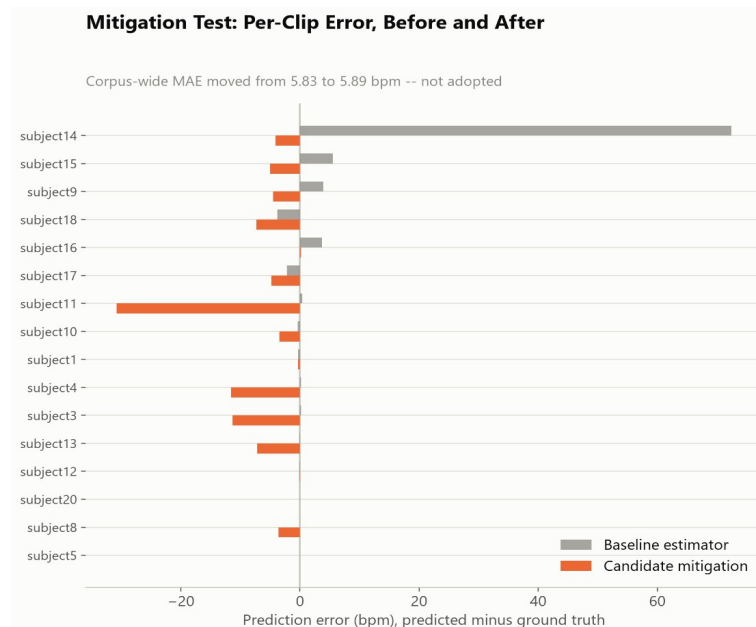


Figure 11. Per-clip prediction error before and after the peak-selection mitigation, sorted by baseline error magnitude. The mitigation fixed subject14 but introduced comparable new errors elsewhere.

This is the second post-hoc estimator-level heuristic tested in this work and rejected on the same evidentiary standard, following the identical rejection of the Welch's-method candidate in Section 4.2. The heuristic was not adopted; plain FFT peak selection remains the reported estimator throughout Sections 5.1-5.3.

6. Discussion

6.1 Interpreting the Reproduction Results

Section 5.1's central result, all four methods converging to under 1.5 bpm MAE on the DATASET_2 test split, is a pass by Section 3.6's own reproduction requirement: the pipeline reproduces published-quality accuracy on the benchmark protocol most published rPPG numbers are measured against. The unresolved PhysNet DATASET_1 regression (46.5 and 43.1 bpm MAE, up from 15.9 and 57.5 bpm before the YuNet swap) is reported rather than hidden because it does not fit this positive result cleanly. The most plausible explanation, not yet directly tested, is that DATASET_1's subjects are already more tightly framed than DATASET_2's, so YuNet's tighter crop may crop too close on already-tight framing; a targeted crop-margin ablation could test this, but this work's timeline did not allow it. We treat this explicitly as an open problem in Section 7 rather than let Section 5.1's headline result imply DATASET_1 is equally well understood.

6.2 What the Defects Imply for Fairness Auditing

Each defect in Section 4.2 was, uncorrected, large enough on its own to fabricate a confident, plausible-looking fairness or robustness finding of exactly the kind this project was designed to detect. Had the CHROM demeaning defect gone uncorrected, a stratified comparison would have reported CHROM as catastrophically unreliable rather than accurate to under one beat per minute, not because CHROM's fairness properties differ across conditions, but because the implementation was broken in a way unrelated to fairness. Had the face-detector defect gone uncorrected, any DATASET_1-versus-DATASET_2 comparison would have reported DATASET_2 subjects as substantially harder for every method simultaneously, easy to over-interpret as a genuine subject property rather than what it actually was: a detection-quality problem affecting a specific subset of clips regardless of subject. This is the project's central methodological argument: a fairness or robustness claim published without this level of implementation scrutiny is not, by default, trustworthy, because the failure modes that would fabricate a false claim are ordinary software defects, not rare or exotic ones.

6.3 Interpreting the Condition-Gap Results

The measured resting-versus-post-exercise gaps (Section 5.3) are directionally consistent across three of four methods, with the fourth (PhysNet, PURE-trained) showing almost no gap. Two caveats bound how much weight this result can bear. First, the after-exercise condition is a single clip ($n=1$): every gap reported here is one data point, not a distribution, and should be read as directional evidence rather than a statistically powered estimate. Second, at least one method's apparent condition gap (PhysNet, UBFC-trained) is substantially driven by a single isolated clip-level failure (subject14, Section 5.4) rather than a broad, condition-wide effect, a distinction invisible without reporting per-clip results alongside the pooled statistic.

6.4 On the Two Negative Mitigation Results

Both candidate mitigations tested in this work, Welch's-method HR estimation (Section 4.2) and lower-peak-preference selection (Section 5.4), followed the same discipline: test against the full relevant corpus, not only the motivating case, and report the result honestly regardless of outcome. Both were rejected. We regard this as a meaningful, not merely a null, finding: it shows the residual per-clip errors in Section 5.3 are not correctable by a global post-hoc estimator rule, because any such rule trades one set of clip-level failures for a different set. This narrows future mitigations toward either model-level fixes (Section 7) or context the current single-window protocol lacks, such as a temporal prior from a neighboring recording window.

6.5 Relationship to the Originally Proposed Skin-Tone Audit

This work did not test the Fitzpatrick skin-tone stratified hypothesis the original proposal targeted (Section 2.5), and the condition-based results in Section 5.3 are not a substitute finding for it; they answer a different question about robustness to physiological and motion state, not demographic fairness. What this work does establish, and what we believe transfers directly to the originally proposed axis, is the methodological infrastructure demonstrated in Sections 4.1, 4.2, and 4.4: byte-level duplicate-data detection, held-out validation of preprocessing decisions, and cryptographic verification of model artifacts. A future skin-tone audit on this same pipeline, once MMPD or EquiPleth access is obtained, would need exactly these same checks; per Section 6.2, it would produce an untrustworthy result without them.

7. Conclusions

This project set out to audit a state-of-the-art rPPG method for both robustness and fairness simultaneously, an intersection the reviewed literature leaves unaudited. Gated data access constraints made the originally proposed skin-tone stratified axis untestable within the project timeline; rather than report an audit that could not honestly be run, we redirected the same standard of rigor toward the axis our obtained data, a pooled, subject-independent 22-clip corpus spanning both public releases of UBFC-rPPG, actually supports, while documenting that redirection transparently rather than presenting it as the original plan.

- A validated, subject-independent, dual-format UBFC-rPPG evaluation pipeline, reproducing four baseline methods (CHROM, POS, and two PhysNet checkpoints) to within 1.5 bpm MAE of published benchmark accuracy on the standard 42-subject protocol.
- Three concrete implementation defects, in CHROM's chrominance normalization, PhysNet's chunk-boundary reconstruction, and the shared face-detection front end, found, root-caused, fixed, and verified with quantitative before/after evidence, each large enough uncorrected to have fabricated a false fairness or robustness finding.
- A resolved, empirically-confirmed diagnosis of an initially paradoxical cross-checkpoint result, illustrating and correcting a specific, generalizable failure mode: evaluating a pretrained model outside its true training domain while believing it to be in-domain.
- A measured, honestly-caveated cross-condition robustness comparison (resting versus post-exercise), including the recognition that one method's apparent condition gap is driven by an isolated per-clip failure rather than a broad effect.
- Two rigorously tested and rejected post-hoc mitigation attempts, reported as negative results because they demonstrate the residual error is not correctable by a global estimator-level rule.
- Two self-identified and corrected gaps in this work's own reporting practice, surfaced by independent review and closed with additional evidence rather than argument: the same standard this project applies to the field it audits.

The unifying conclusion across all six contributions is that reproducibility rigor is not a preliminary step separate from fairness and robustness auditing in rPPG research; it is a precondition for that auditing producing a trustworthy answer at all. A stratified accuracy comparison computed on top of an unvalidated pipeline is not evidence about fairness or robustness; it is, at best, unlabeled evidence about implementation quality wearing a fairness result's clothing. We hope the specific defects, checks, and negative results documented here function as a directly reusable checklist for the skin-tone audit this project originally set out to run, and for any rPPG fairness or robustness claim that follows.

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Appendix A: Full Per-Clip Results

Ground truth and predicted heart rate (bpm) for every clip in the pooled 22-clip corpus, final validated pipeline. "D1"/"D2" denote dataset family.

Subject	Fam.	GT	CHROM	POS	PhysNet-UBFC	PhysNet-PURE
10-gt	D1	70.1	71.0	71.0	138.9	70.8
11-gt	D1	78.7	76.5	79.0	79.1	159.1
5-gt	D1	76.5	76.5	76.5	147.7	154.2
6-gt	D1	84.3	84.5	84.5	166.6	181.9
8-gt	D1	62.4	58.8	58.8	118.7	62.8
after-exercise	D1	105.6	104.1	104.5	105.9	103.7
subject1	D2	107.5	107.2	107.2	107.2	107.2
subject10	D2	119.5	120.0	119.5	119.1	119.5
subject11	D2	127.0	87.1	127.1	127.4	127.0
subject12	D2	65.2	65.1	65.1	65.0	65.0
subject13	D2	109.6	109.6	109.1	109.4	109.4
subject14	D2	75.7	70.6	80.9	148.1	75.6

Subject	Fam.	GT	CHROM	POS	PhysNet-UBFC	PhysNet-PURE
subject15	D2	111.0	117.1	58.5	116.5	116.5
subject16	D2	92.1	92.6	92.6	95.8	95.8
subject17	D2	87.0	84.7	89.1	84.8	84.8
subject18	D2	127.7	123.8	124.3	123.9	123.9
subject20	D2	131.8	131.5	131.5	131.8	131.8
subject3	D2	98.7	87.5	82.3	98.9	98.9
subject4	D2	110.5	110.7	105.5	110.7	110.7
subject5	D2	99.7	100.0	99.6	99.8	99.8
subject8	D2	111.2	111.3	110.8	111.2	111.2
subject9	D2	105.6	105.2	70.1	109.4	115.1

Appendix B: Subject-Independent Split Manifest

The pooled 22-clip corpus, split by subject (never by frame). Test-split subjects fall entirely within DATASET_2.

Subject ID	Dataset Family	Split
10-gt	DATASET_1	train
11-gt	DATASET_1	train
5-gt	DATASET_1	train
6-gt	DATASET_1	train
8-gt	DATASET_1	train
after-exercise	DATASET_1	train
subject11	DATASET_2	train
subject14	DATASET_2	train
subject15	DATASET_2	train
subject16	DATASET_2	train
subject18	DATASET_2	train
subject20	DATASET_2	train
subject3	DATASET_2	train
subject4	DATASET_2	train
subject8	DATASET_2	train
subject10	DATASET_2	val
subject12	DATASET_2	val
subject9	DATASET_2	val
subject1	DATASET_2	test
subject13	DATASET_2	test
subject17	DATASET_2	test
subject5	DATASET_2	test

Appendix C: Software and Tools

Python 3.11; OpenCV 4.10 (Haar cascade, superseded by YuNet, cv2.FaceDetectorYN); NumPy; SciPy; pandas; PyTorch (CPU inference); rPPG-Toolbox [9] pretrained PhysNet checkpoints (UBFC-rPPG-trained and PURE-trained, DiffNormalized configuration) and reference data-loader source (consulted directly to determine checkpoint training-data provenance, Section 4.3); YuNet face detector model (face_detection_yunet_2023mar.onnx, OpenCV Zoo). All preprocessing, baseline, evaluation, and figure-generation code is organized as a versioned pipeline: preprocessing/rppg_preprocess.py (dataset discovery, ROI extraction, ground-truth alignment, subject-independent split), preprocessing/baselines_classical.py (CHROM, POS), preprocessing/physnet_infer.py (PhysNet inference), preprocessing/step5_condition_analysis.py (cross-condition evaluation), and preprocessing/make_figures.py (all figures in this report).